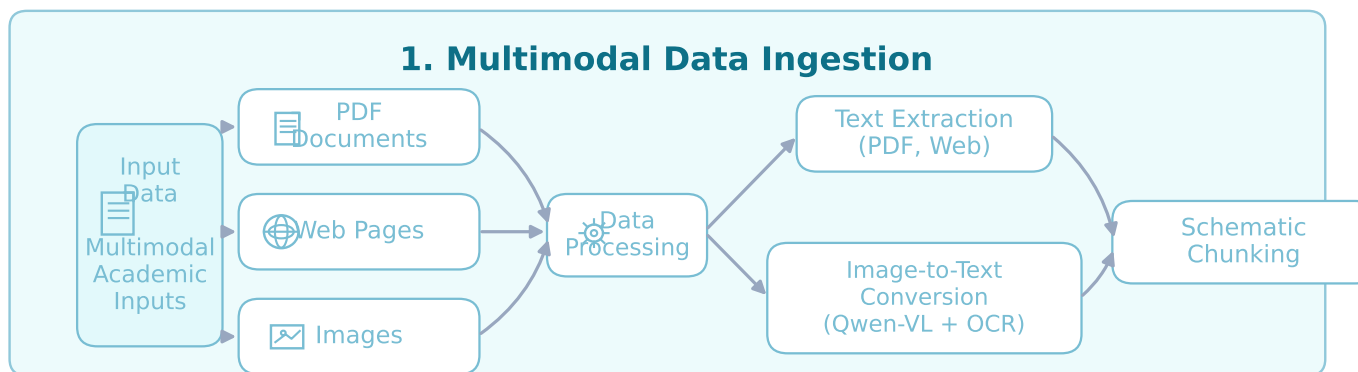
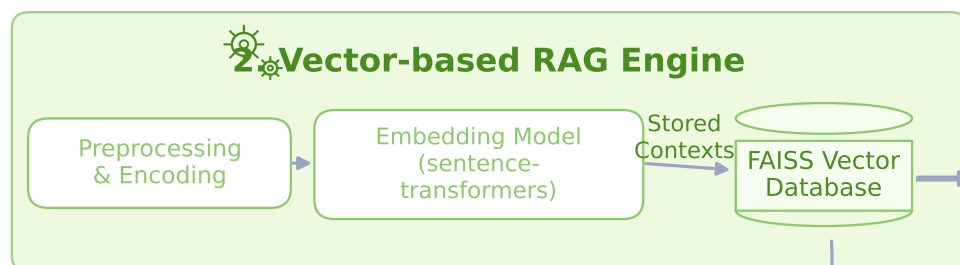


# Publication-oriented architecture of the offline academic assistant pipeline

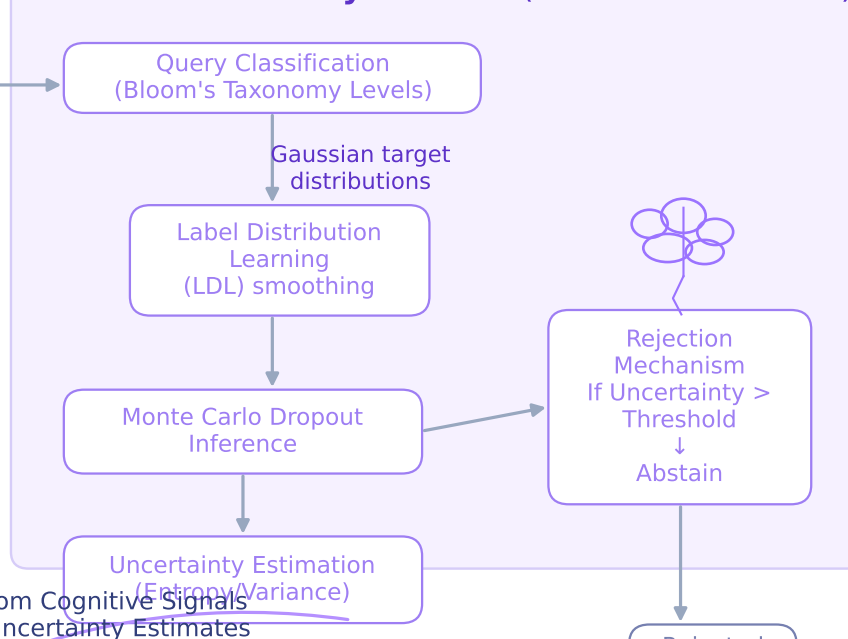
## 1. Multimodal Data Ingestion



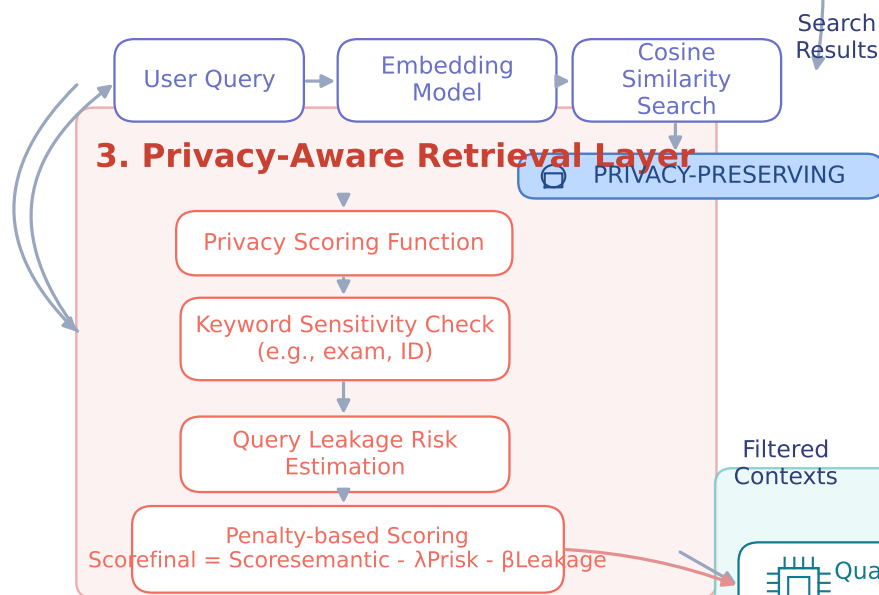
## 2. Vector-based RAG Engine



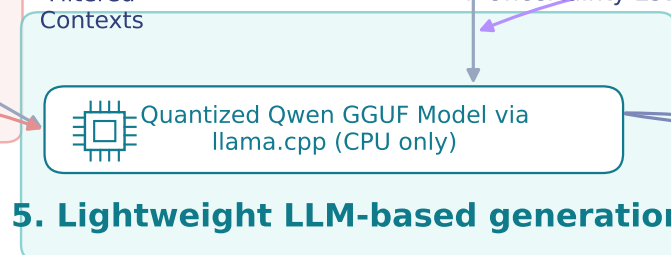
## 4. LDL Uncertainty Module (Monte Carlo based)



## 3. Privacy-Aware Retrieval Layer



## 5. Lightweight LLM-based generation



Bloom Cognitive Signals + Uncertainty Estimates